Quiz on estimates of quantities related to the electrical conductivity problem

* Required

* This form will record your name, please fill your name.

Make a Choice

1

Choose a path *

- ◯ Estimates of microscopic quantities (easy but more questions)
- ◯ Estimates of macroscopic quantities (moderate but less questions)

Estimates of microscopic quantities

2

At room temperature, order in eV is *

- The average kinetic energy $\langle T \rangle$ at

$$\langle K \rangle = \frac{3}{2} k_B$$

where k_B is the Boltzmann constar

- ○ 10^-1
- ○ 10^-2
- ○ 10^-3
- ○ 10^-4

3

At room temperature, order in ms^-1 is *

Definition
Thermal velocity v_{th} is defined as the average ve
electron.

- 10^4
- 10^5
- 10^6
- 10^2

4

Estimate order of electrical field in V/m inside the copper wire *

Question

A copper wire of length $L = 2\,\mathrm{m}$ and diameter $d{=}1$
across a DC source. When a voltage of $V{=}0.5\,\mathrm{V}$ is
ends of the wire, an electric current of $I{=}22\,\mathrm{A}$ is ob
measurements, find the electrical conductivity σ of

- ○ 10^0
- ○ 10^1
- ○ 10^-1
- ○ 10^+2

5

Estimate order of condutivity in Sm^-1 *

2. Electrical conductivity

A meteorite has fallen from outer space near Vignan Univers
assigned you to find the type of material. It is difficult to mal
you cut a cube out of the material and perform electrical con
with the following observations.
Side length of cube = 10 cm
Opposite faces are connected across a DC source. When a vol
applied, an electric current of I=1 nA (n for nano) is observed
What is the conductivity of the material? Which type of mate

- 10^-7
- 10^+7
- 10^-12
- 10^-10

6

Estimate drift velocity in ms^-1. *

3. For Cu, the experimental observed $n = 8.47 \times 10$

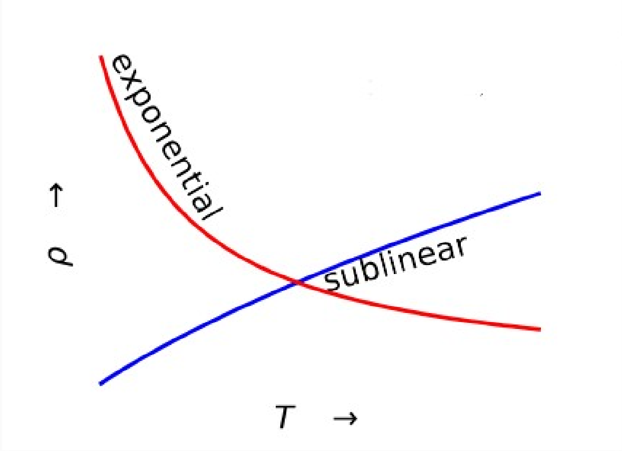
Question

A copper wire of length $L = 2\,\mathrm{m}$ and diameter $d = 1$
across a DC source. When a voltage of $V = 0.5\,\mathrm{V}$ is
ends of the wire, an electric current of $I = 22\,\mathrm{A}$ is ob

- 10^-2
- 10^-3
- 10^+2
- 10^+3

7

Metals behave as *



- Red curve
- Blue curve
- Red or Blue
- None of the above

Estimates of macroscopic quantities

8

*

Amount of charge in C required

1 g of H_2 gas from electrolysis of

order of

- 10^+3
- 10^+4
- 10^+5
- 10^+6

9

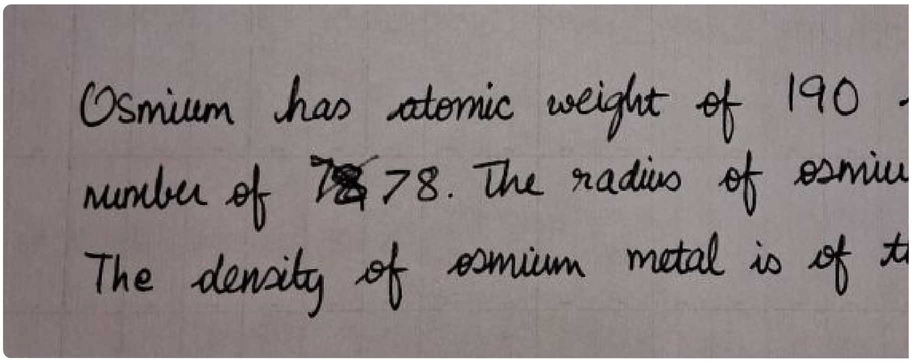
Estimate of surface density *

Samsung A14 display is 16.7 cm X
resolution is 1920 x 1080 pixels. If t
radius of Si atom is 4Å, then th
atoms that are seen in one pixel
the order of

- 10^8
- 10^9
- 10^10
- 10^11

10

Estimate of number density *



Osmium has atomic weight of 190
number of ~~78~~ 78. The radius of osmiu
The density of osmium metal is of t

- 10^0
- 10^1
- 10^3
- 10^4

11

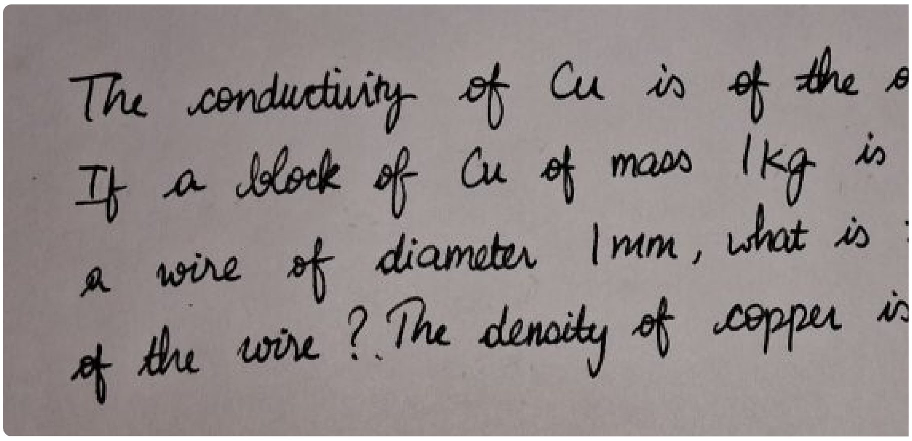
*

The relaxation time in copper is
order of 10 fs. If the electric fi
across a wire made out of copper
then the drift velocity attained by
of the order of (in ms^{-1})

- 10^-1
- 10^-2
- 10^+1
- 10^+2

12

*

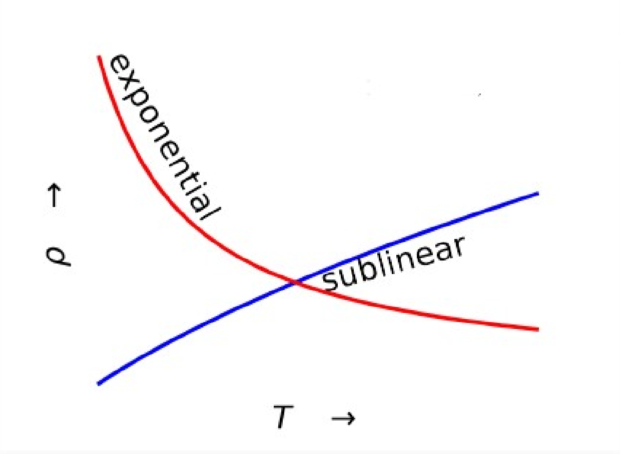


The conductivity of Cu is of the
If a block of Cu of mass 1kg is
a wire of diameter 1mm, what is
of the wire? The density of copper is

- ◯ 10^0
- ◯ 10^1
- ◯ 10^2
- ◯ 10^3

13

Metals behave as *



- Red curve
- Blue curve
- Red of Blue curve
- None of the above